

# CS336 Assignment 1 Basics Writeup

## Problem (unicode1): Understanding Unicode

- (a) `chr(0)` returns the Null character (U+0000).
- (b) The string representation (`__repr__()`) displays the escaped hex code `'\x00'`, whereas the printed representation outputs the actual invisible control character.
- (c) When this character occurs in a Python string, it behaves as a normal character (increasing length without terminating the string) but is typically invisible when printed to the console.

## Problem (unicode2): Unicode Encodings

- (a) UTF-8 is preferred because it is variable-width and highly space-efficient for ASCII characters (using 1 byte vs 2-4 bytes in UTF-16/32), which dominate most training corpora (like English text and code), and it avoids the issue of null-byte sparsity found in fixed-width encodings.
- (b) "A 中 😊". The function fails to decode the bytes correctly because it does not handle multi-byte characters properly. It attempts to decode each byte as a separate character, which can lead to errors or unexpected results.
- (c) `"\xe4\x00"`. The sequence starts with `"\xe4"` (11100100), implicating that it is a 3-byte sequence. However, the second byte is `"\x00"` (00000000) instead of `"10xxxxxx"`, so it is not a valid UTF-8 encoding of a character, and there are only two bytes available to decode.

## Problem (train\_bpe): BPE Tokenizer Training

Implemented `pretokenize`, `init_vocab`, `build_words`, `init_pairs`, `merge`.

## Problem (train\_bpe\_tinystories): BPE Training on TinyStories

- (a) 320s in 4vcpu, 32s in 128vcpu, both took up about 3.4 GB RAM. Longest token (ID 7160): `b' accomplishment'`.
- (b) Pretokenize took the most time.

## Problem (train\_bpe\_expts\_owt): BPE Training on OpenWeb-Text

- (a) 320s in 4vcpu, 32s in 128vcpu, both took up about 3.4 GB RAM. Longest token (ID 7160): `b' accomplishment'`.
- (b) Pretokenize took the most time.